MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.1: Orthogonality and orthogonal bases

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Recall from Vectors that the dot product is symmetric and distributes over linear combinations, by Properties of the dot product. The norm satisfies $\|\mathbf{x}\|^2 = \mathbf{x} \cdot \mathbf{x}$, as noted in the referenced item. Two vectors are orthogonal when their dot product is zero. A nonzero vector can be normalized by dividing it by its norm.

Orthogonal sets and the expansion theorem

We now study collections of mutually orthogonal vectors and how they provide convenient bases and coordinate formulas.

Definition: Orthogonal and orthonormal sets

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is called an orthogonal set if the set does not contain the zero vector, and $\mathbf{x}_i \cdot \mathbf{x}_j = 0$ for $i \neq j$. A set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is called orthonormal if it is orthogonal and if in addition, $\|\mathbf{x}_i\| = 1$ for each i.

Note that $\{\mathbf{x}\}$ is an orthogonal set if $\mathbf{x} \neq \mathbf{0}$. The standard basis $\{\mathbf{e}_1, \cdots, \mathbf{e}_n\}$ (see Standard basis) is an important orthonormal set in $\mathbb{R}^n$.

Example: Checking orthogonality and normalizing vectors (U4-LO1)

Consider the following vectors in $\mathbb{R}^4$, written as column vectors:

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

Verify that these vectors form an orthogonal set in $\mathbb{R}^4$. Then find the orthonormal set obtained by 'normalizing' these vectors, i.e., multiplying each vector by an appropriate constant to obtain a unit vector.

To verify that $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ form an orthogonal set, we need to check that $\mathbf{f}_i \cdot \mathbf{f}_j = 0$ for all $i \neq j$. We calculate that

$$\begin{aligned} \mathbf{f}_1 \cdot \mathbf{f}_2 &= 1 \cdot 1 + 1 \cdot 0 \\ &\quad + 1 \cdot 1 + (-1) \cdot 2 = 1 + 0 + 1 - 2 = 0 \end{aligned}$$

$$\begin{aligned} \mathbf{f}_1 \cdot \mathbf{f}_3 &= 1 \cdot (-1) + 1 \cdot 0 \\ &\quad + 1 \cdot 1 + (-1) \cdot 0 = -1 + 0 + 1 + 0 = 0 \end{aligned}$$

$$\begin{aligned} \mathbf{f}_1 \cdot \mathbf{f}_4 &= 1 \cdot (-1) + 1 \cdot 3 \\ &\quad + 1 \cdot (-1) + (-1) \cdot 1 \\ &= -1 + 3 - 1 - 1 = 0 \end{aligned}$$

$$\mathbf{f}_2 \cdot \mathbf{f}_3 = 1 \cdot (-1) + 0 \cdot 0 + 1 \cdot 1 + 2 \cdot 0 = -1 + 0 + 1 + 0 = 0$$

$$\mathbf{f}_2 \cdot \mathbf{f}_4 = 1 \cdot (-1) + 0 \cdot 3 + 1 \cdot (-1) + 2 \cdot 1 = -1 + 0 - 1 + 2 = 0$$

$$\mathbf{f}_3 \cdot \mathbf{f}_4 = (-1) \cdot (-1) + 0 \cdot 3 + 1 \cdot (-1) + 0 \cdot 1 = 1 + 0 - 1 + 0 = 0.$$

Since all these dot products are zero, the set is orthogonal.
To normalize these vectors, we need to calculate their lengths and divide each vector by its length. We find that

$$\|\mathbf{f}_1\| = \sqrt{1^2 + 1^2 + 1^2 + (-1)^2} = \sqrt{4} = 2$$
$$\|\mathbf{f}_2\| = \sqrt{1^2 + 0^2 + 1^2 + 2^2} = \sqrt{6}$$
$$\|\mathbf{f}_3\| = \sqrt{(-1)^2 + 0^2 + 1^2 + 0^2} = \sqrt{2}$$
$$\|\mathbf{f}_4\| = \sqrt{(-1)^2 + 3^2 + (-1)^2 + 1^2} = \sqrt{12} = 2\sqrt{3}.$$

Therefore the normalized vectors are given by:

$$\frac{1}{\|\mathbf{f}_1\|}\mathbf{f}_1 = \frac{1}{2}\mathbf{f}_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$$\frac{1}{\|\mathbf{f}_2\|}\mathbf{f}_2 = \frac{1}{\sqrt{6}}\mathbf{f}_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ 0 \\ \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$\frac{1}{\|\mathbf{f}_3\|}\mathbf{f}_3 = \frac{1}{\sqrt{2}}\mathbf{f}_3 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$

$$\frac{1}{\|\mathbf{f}_4\|}\mathbf{f}_4 = \frac{1}{2\sqrt{3}}\mathbf{f}_4 = \begin{bmatrix} -\frac{1}{2\sqrt{3}} \\ \frac{3}{2\sqrt{3}} \\ -\frac{1}{2\sqrt{3}} \\ \frac{1}{2\sqrt{3}} \end{bmatrix}.$$

Theorem: Pythagoras' theorem

If $\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_k\}$ is an orthogonal set in $\mathbb{R}^n$, then

$$\|\mathbf{f}_1 + \mathbf{f}_2 + \cdots + \mathbf{f}_k\|^2 = \|\mathbf{f}_1\|^2 + \|\mathbf{f}_2\|^2 + \cdots + \|\mathbf{f}_k\|^2.$$

Theorem

Every orthogonal set in $\mathbb{R}^n$ is linearly independent.

Why is this true?

Let $\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_k\}$ be an orthogonal set in $\mathbb{R}^n$. To show the set is linearly independent (recalling Linear Independence), we must show the only solution to the linear equation

$$t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k = \mathbf{0}$$

occurs when $t_1 = \cdots = t_k = 0$.
If we take the dot product of both sides of the equation with $\mathbf{f}_1$, we find that

$$\mathbf{f}_1 \cdot (t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_k\mathbf{f}_k) = \mathbf{f}_1 \cdot \mathbf{0}$$

and therefore that

$$t_1(\mathbf{f}_1 \cdot \mathbf{f}_1) + t_2(\mathbf{f}_1 \cdot \mathbf{f}_2) + \cdots + t_k(\mathbf{f}_1 \cdot \mathbf{f}_k) = 0$$

Since the set of vectors is orthogonal, $\mathbf{f}_1 \cdot \mathbf{f}_j = 0$ for all $j \neq 1$, and $\mathbf{f}_1 \cdot \mathbf{f}_1 = \|\mathbf{f}_1\|^2$, so we conclude that

$$t_1\|\mathbf{f}_1\|^2 = 0.$$

Since $\mathbf{f}_1 \neq \mathbf{0}$ (by definition, an orthogonal set cannot contain the zero vector), $\|\mathbf{f}_1\|^2 > 0$, and so it must be true that $t_1 = 0$. But now repeating this argument with 1 replaced by any index i, we conclude that $t_i = 0$. Therefore the set $\{\mathbf{f}_1, \ldots, \mathbf{f}_k\}$ is linearly independent.

Theorem: Expansion theorem

Let $\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_m\}$ be an orthogonal basis of a subspace U of $\mathbb{R}^n$. If $\mathbf{v}$ is any vector in U, we have

$$\mathbf{v} = \left(\frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\right)\mathbf{f}_1 + \left(\frac{\mathbf{v} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2}\right)\mathbf{f}_2 + \cdots + \left(\frac{\mathbf{v} \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2}\right)\mathbf{f}_m$$

Why is this true?

Since $\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_m\}$ spans U, any $\mathbf{v} \in U$ can be written uniquely as

$$\mathbf{v} = t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_m\mathbf{f}_m$$

for some $t_1, \ldots, t_m$. To find coefficient t_1, take the dot product of both sides with $\mathbf{f}_1$, obtaining that

$$\begin{aligned}\mathbf{v} \cdot \mathbf{f}_1 &= (t_1\mathbf{f}_1 + t_2\mathbf{f}_2 + \cdots + t_m\mathbf{f}_m) \cdot \mathbf{f}_1 \\ &= t_1(\mathbf{f}_1 \cdot \mathbf{f}_1) + t_2(\mathbf{f}_2 \cdot \mathbf{f}_1) + \cdots + t_m(\mathbf{f}_m \cdot \mathbf{f}_1).\end{aligned}$$

Since the basis is orthogonal, $\mathbf{f}_i \cdot \mathbf{f}_j = 0$ for $i \neq j$, and so the equation simplifies to

$$\mathbf{v} \cdot \mathbf{f}_1 = t_1\|\mathbf{f}_1\|^2.$$

Rearranging, we conclude that

$$t_1 = \frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}$$

A similar argument shows that

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

for each i, concluding the proof.

Definition: The Fourier expansion in an orthogonal basis

Given an orthogonal basis $\{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ for some subspace V of $\mathbb{R}^n$, the expansion of $\mathbf{v} \in V$ as a linear combination of this basis is called

the Fourier expansion of the vector $\mathbf{v}$, and the coefficients

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

are called the Fourier coefficients of $\mathbf{v}$.

Activity: Expansion with two orthogonal vectors (U4-LO1)

Consider the vectors

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{f}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

in $\mathbb{R}^3$, and let $U = \text{span}\{\mathbf{f}_1, \mathbf{f}_2\}$.

1. Verify that $\{\mathbf{f}_1, \mathbf{f}_2\}$ is an orthogonal basis of U.
2. Use the Fourier coefficient formula to find the expansion of $\mathbf{v}$ in this basis. Check by direct addition that your linear combination equals $\mathbf{v}$.

Both basis vectors are nonzero, and

$$\mathbf{f}_1 \cdot \mathbf{f}_2 = 1 \cdot 1 + 1 \cdot (-1) + 0 \cdot 1 = 0.$$

Thus the set is orthogonal and therefore linearly

independent. It spans U by definition, so it is an orthogonal basis of U.
The squared lengths are

$$\|\mathbf{f}_1\|^2 = 1^2 + 1^2 + 0^2 = 2,$$
$$\|\mathbf{f}_2\|^2 = 1^2 + (-1)^2 + 1^2 = 3.$$

The Fourier coefficients are therefore

$$t_1 = \frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} = \frac{3 \cdot 1 + 1 \cdot 1 + 1 \cdot 0}{2} = \frac{4}{2} = 2,$$
$$t_2 = \frac{\mathbf{v} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} = \frac{3 \cdot 1 + 1 \cdot (-1) + 1 \cdot 1}{3} = \frac{3}{3} = 1.$$

Checking the resulting expansion gives

$$2\mathbf{f}_1 + \mathbf{f}_2 = 2\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \mathbf{v}.$$

This also verifies that $\mathbf{v} \in U$.

Activity: Expansion in an orthogonal basis (U4-LO1)

Let $\mathbf{v} = (a, b, c, d)$ be a vector in $\mathbb{R}^4$. Find the Fourier expansion of $\mathbf{v}$ as a linear combination of the orthogonal basis $\{\mathbf{f}_1, \mathbf{f}_2, \mathbf{f}_3, \mathbf{f}_4\}$ given in Checking orthogonality and normalizing vectors (U4-LO1). Recall that

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix} \quad \mathbf{f}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{f}_4 = \begin{bmatrix} -1 \\ 3 \\ -1 \\ 1 \end{bmatrix}.$$

We need to find the Fourier coefficients

$$t_i = \frac{\mathbf{v} \cdot \mathbf{f}_i}{\|\mathbf{f}_i\|^2}$$

for $i = 1, 2, 3, 4$. In Checking orthogonality and normalizing vectors (U4-LO1) we already computed the lengths of the elements of the basis, namely, that

$$\|\mathbf{f}_1\| = 2, \qquad \|\mathbf{f}_2\| = \sqrt{6},$$
$$\|\mathbf{f}_3\| = \sqrt{2} \quad \text{and} \quad \|\mathbf{f}_4\| = 2\sqrt{3}.$$

We now calculate the dot products:

$$\mathbf{v} \cdot \mathbf{f}_1 = a \cdot 1 + b \cdot 1 + c \cdot 1 + d \cdot (-1) = a + b + c - d$$

$$\mathbf{v} \cdot \mathbf{f}_2 = a \cdot 1 + b \cdot 0 + c \cdot 1 + d \cdot 2 = a + c + 2d$$

$$\mathbf{v} \cdot \mathbf{f}_3 = a \cdot (-1) + b \cdot 0 + c \cdot 1 + d \cdot 0 = -a + c$$

$$\mathbf{v} \cdot \mathbf{f}_4 = a \cdot (-1) + b \cdot 3 + c \cdot (-1) + d \cdot 1 = -a + 3b - c + d$$

Therefore, the Fourier coefficients are:

$$t_1 = \frac{\mathbf{v} \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2} = \frac{a+b+c-d}{4}$$
$$t_2 = \frac{\mathbf{v} \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2} = \frac{a+c+2d}{6}$$
$$t_3 = \frac{\mathbf{v} \cdot \mathbf{f}_3}{\|\mathbf{f}_3\|^2} = \frac{-a+c}{2}$$
$$t_4 = \frac{\mathbf{v} \cdot \mathbf{f}_4}{\|\mathbf{f}_4\|^2} = \frac{-a+3b-c+d}{12}$$

The Fourier expansion of $\mathbf{v}$ in the basis $\{\mathbf{f}_1, \ldots, \mathbf{f}_4\}$ is therefore

$$\mathbf{v} = \left(\frac{a+b+c-d}{4}\right)\mathbf{f}_1 + \left(\frac{a+c+2d}{6}\right)\mathbf{f}_2 + \left(\frac{-a+c}{2}\right)\mathbf{f}_3 + \left(\frac{-a+3b-c+d}{12}\right)\mathbf{f}_4.$$

Dot products and orthogonality now become tools for finding closest vectors.